

Task-1: Introduction to automobile and electric vehicles

1. Automobile system explanation

A Comprehensive Report on Core Automotive Systems: Gas vs. Electric

1. Foundation Automobile Systems

Every vehicle on the road relies on a core group of mechanical systems. These systems handle the weight of the car, keep the ride smooth, and give the driver control.

- **What is automobile?**

An **automobile is a complex self-propelled vehicle** composed of multiple interconnected mechanical, electrical, and electronic systems that work in tandem to safely transport passengers and goods. These systems work cohesively to generate power, transmit it to the wheels, and allow precise driver control.

- **The Chassis and Suspension**

The chassis is the vehicle's structural backbone. Most modern passenger cars use a unibody design. This means the body and frame are built as one single piece. This setup keeps the vehicle lightweight and includes built-in crumple zones to protect passengers during a crash.

The suspension connects this frame to the wheels. It uses a mix of springs and shock absorbers to damp road vibrations. This keeps the cabin comfortable while forcing the tires to stay flat against the pavement for optimal grip.

- **Steering and Braking Controls**

Drivers change direction using a steering system, usually a rack-and-pinion setup. Turning the steering wheel moves a mechanical rack left or right, which pivots the front tires. Most modern vehicles use electronic power steering to make turning easier at low speeds.

The brakes rely on hydraulic fluid to slow the car down. Pressing the brake pedal forces fluid through lines to clamp heavy pads against spinning metal discs at the wheels. The resulting friction changes kinetic energy into heat. An Anti-lock Braking System (ABS) pulses the brakes automatically during a hard stop to prevent the tires from skidding.

2. Electric Vehicle (EV) Components and Working Principles

Electric vehicles swap out hundreds of moving mechanical parts for a simplified system driven by software, chemistry, and magnets.

- **The Traction Battery Pack**The battery pack sits flat under the floor of the vehicle to keep the center of gravity low. It is built from thousands of individual lithium-ion cells wired together.
- **Cell Monitoring:** The Battery Management System (BMS) checks cell voltages constantly.
- **Safety Control:** The software stops the battery from overcharging.
- **Thermal Management:** Liquid coolant pumps through the pack to stop overheating.

The Power Inverter

The battery stores energy as Direct Current (DC), but the electric motor requires Alternating Current (AC) to spin. The power inverter serves as the translator between the two.

- **Current Conversion:** It changes DC battery power into three-phase AC power.
- **Speed Management:** It adjusts the electricity's frequency to control vehicle speed.
- **Auxiliary Power:** A built-in converter drops high voltage down to 12V for headlights and screens.

The Electric Motor and Gearbox

Electric traction motors use alternating current to create a rotating magnetic field. This field pulls an internal rotor, causing it to spin and turn the drive axles.

- **Instant Torque:** The motor delivers maximum pulling power immediately from zero RPM.
- **Simplified Gearing:** It runs efficiently across a massive RPM range.
- **No Shifting:** The car uses a simple single-speed reduction gear instead of a multi-gear transmission.

2.4 Regenerative Braking

When you take your foot off the accelerator pedal, the electric motor reverses its role. The rolling momentum of the car spins the motor, turning it into an electrical generator. This action slows the vehicle down naturally while sending electricity back into the battery pack, recovering energy that gas cars waste as heat.

3. Direct Technical Comparison

Feature	Gas Cars (ICE)	Electric Vehicles (EV)
Fuel Type	Gasoline or Diesel	Electricity
Core Mechanism	Internal Combustion Engine	Electromagnetic Motor
Energy Efficiency	Low (around 30% useful energy)	High (around 90% useful energy)

Power Band	Requires RPM build-up	Instantaneous from a stop
Moving Parts	Thousands of internal components	Roughly twenty moving components
Transmission	6 to 10 gears required	Single-speed fixed gear
Tailpipe Waste	Carbon dioxide and pollutants	None
Refueling Time	3 to 5 minutes	15 minutes to several hours

4. Advantages and Real-World Challenges

Major Advantages

- **Lower Operating Costs:** Electricity costs less per mile than gasoline.
- **Simpler Maintenance:** No oil changes, spark plugs, or timing belts exist to break down.
- **Better Handling:** The heavy under-floor battery prevents the car from leaning in corners.
- **Zero Tailpipe Emissions:** The powertrain produces no direct exhaust pollution.

4.2 Current Challenges

- **Unreliable Infrastructure:** Public fast-charging stations suffer from frequent downtime and broken plugs.
- **Winter Range Loss:** Freezing weather slows down battery chemistry and drops range significantly.
- **Higher Purchase Price:** High raw material costs make EVs more expensive upfront than gas equivalents.
- **Supply Chain Impact:** Sourcing materials like cobalt and lithium causes localized environmental damage.

